

4.c) Discuss the importance of immunological and biochemical methods in the detection of genetic disorders. (15 Marks)

Immunological Methods are the one which is fighting against the pathogen on the basis of antigen-antibody mechanism or cell mediated immunity.

Biochemical Methods are the one which used to analyse the cells substrate and its component like DNA, RNA or protein.

Importance of Immunological method in detection of Genetic Disorders

① HLA genes are the hallmark of immunity and it prevents from autoimmune genetic disorder.

② Any disorder in HLA B27 → Spondylitis, HLA DR3 & HLA DR4

Causes the Type I diabetes.

③ Body will produce antibodies against our own cells and it can be screened out to know, which disorder is present.

④ Each Antibody has specific paratope for their epitope called Monoclonal Antibody.

⑤ The Cell Mediated T cells or T helper cells → when grow in more numbers → we can analyse the disorder.

Biochemical methods in detection of Genetic disorders by :-

① By the process Chromatography the DNA fragments are cut into pieces and then analysed.

② In the gel electrophoresis method, they are auto-radiographed to look into any m-RNA/probe of the mutated gene is present or not.

③ It further help in arranging the fragment in increasing or decreasing order. to check the range of mutation.

↳ whether it is frameshift

↳ whether it is single point

↳ or any large aberrations.

Uses → HLA genes are analysed
→ Gamma factors are analysed
→ blood serums are also analysed

Hence, these methods are used in the analysis of the genetic disorder in human.